

Machine Learning I

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1. Introduction to ML

1.1. Basic definitions.

Object (x).

The thing we are analysing.

Object set ($\mathbb{X}$).

The set of all the possible objects of a certain task. Eucludic space.

Target varibale (y).

The answer to the task we are solving.

Target set ($\mathbb{Y}$).

The set of all the possible targets of a certain task. Eucludic space.

Teaching set ($X := (x_i, y_i)_i$).

A set of object for which there are **correct** targets, and $|X| = \ell$.

Datadrift.

We say we have encountered datadrift when the teaching set is no longer relevant for today or tomorrow.

Factors.

The characteristics of objects x , where d is the amount of factors.

$$X_i = (x_{i1}, \dots, x_{id})$$

Numerical: $x_j \in \mathbb{R}$,

Categorical: $x_j \in \{C_1, \dots, C_m\}$, (Only has the comparison operation)

Ordered: $x_j \in \{C_1, \dots, C_m\}$ (Only has the ordering operation)

1.2. Types of tasks in machine learning.

The type of task in machine learning is defined by the target variable y from the target variable set $\mathbb{Y}$.

Supervised Learning.

The teaching set $X := (x_i, y_i)_i$ for which the targets are aviliable.

1. Linear Regression ($\mathbb{Y} = \mathbb{R}$)

2. Classification ($|\mathbb{Y}| < \infty$) :

(Binary : $\mathbb{Y} = \{0, 1\}$, Multi-class : $\mathbb{Y} = \{1, \dots, k\}$, Multi-label: $\mathbb{Y} = \{0, 1\}^k$)

3. Ranking (Ранжирование)

Unsupervised learning.

We are given a teaching set $X_i := (x_i, y_i)_i$ without the target variables as answers.

1. Clusterization: We try to break a set into subsets such that are similar to each other.
2. Ozenivanie plotnoist: (Search for anomalies)

Partial Learning A method that combines supervised and unsupervised learning.

1.3. Building an algorithm (hypothesis)

The **task** that we will be solving is:

$$\alpha : \mathbb{X} \rightarrow \mathbb{Y}$$

Error function. We need to be able to formally evaluate the quality of algorithms.

As a popular example we have MSE (Mean Squared Error).

$$\frac{1}{\ell} \sum_{i=1}^{\ell} (\alpha(x_i) - y_i)^2$$

The smaller the amount of the error function, the better the algorithm.

Loss function.

$$L : \mathbb{Y} \times \mathbb{Y} \rightarrow \mathbb{R}_+$$

Continuing with the example from before:

$$L(y, z) = (y - z)^2$$

Algorithm ($\mathcal{A}$)

Once we have fixed an error function, we are able to build the algorithm $\alpha(x)$. Beforehand we fix a family of algorithms $\mathcal{A}$.

$$\mathcal{A} = \{\alpha(x) = w_0 + w_1x_1 + \dots + w_dx_d \mid w_0, w_1, \dots, w_d \in \mathbb{R}\}, x_i : i \text{ factor of the object } x$$

$$\alpha(x) = w_0 + w_1x_1 + \dots + w_dx_d = w_0 + \sum_{j=1}^d w_jx_{ij}$$

The best algorithm we are able to obtain from this is obtained through solving the following problem:

$$\frac{1}{\ell} \sum_{j=1}^{\ell} \left(w_0 + \sum_{j=1}^d w_jx_{ij} - y_i \right)^2 \rightarrow \min_{w_0, \dots, w_d}, x_{ij} \text{ is the } j \text{ factor of on an } i \text{ object}$$

The process of finding an optimal algorithm is called training a model (algorithm).

1.4. Filtering data

For an algorithm to be useful we need to be able to feed it good data. Data might need to be altered in order to be considered “good”.

1. Linear models:

Normalize them by subtracting the average,
and dividing by the Var of all columns of the 'objects-factors' matrix.

2. Outlier:

Objects that are not correct examples, & the presence of which ruins a model.

3. Noise:

Objects that do not have any relation to the target variable y .

Overfitting. Is the result of a model that corresponds too closely to the teaching set $X_i := (x_i, y_i)_i$ and fails to fit and / or predict future observations.

As an example:

$$\alpha(x) = \begin{cases} y_i, & x = x_i \\ 0, & x \notin X \end{cases}$$

This would be an awful model, as it would not learn anything at all,
as it follows the target way too closely.

2. Linear models

2.1. Introducing linear models.

Linear models.

We can describe the vast majority of linear regression models as the sum of factors with certain weights.

$$\alpha(x) := w_0 + \sum_{j=1}^d w_j x_j$$

The **parameters** of a model are its **weights (coefficients)** w_j , where w_0 is the bias (free coefficient).

Our form of presenting the linear model can be rewritten as:

$$\alpha(x) := w_0 + \langle w, x \rangle, \quad w = (w_1, \dots, w_d)$$

If we add to each factor description of each object an $(d + 1)$ factor equal to one, it would could as the free coefficient, and would permit us to rewrite the linear model as:

$$\alpha(x) := \langle w, x \rangle$$

2.2. Usage of linear models.

An important criteria to consider is that in the linear model each factor has a linear effect on the target variable, and this target variable does not depend from a combination of factors.

Categorical factors.

Suppose we have a categorical factor the value of which we cannot compare the size of, subtract or add from. These types of factors cannot be used in linear models without additional modification.

One-hot coding.

Essentially, one hot encoding is a method for converting categorical variables into a binary format, where we create new columns for each category where 1 is the category being present and 0 being absent.

Suppose that the categorical factor $f_{j(x)}$ takes in values from the set $C = \{c_1, \dots, c_m\}$. We replace this categorical factor $f_{j(x)}$ for the m binary factors $b_1(x), \dots, b_m(x)$, where each one is the indicator function of one of the possible categorical values.

$$b_i(x) = [f_{j(x)} = c_i], \quad \text{s.t. } b_1(x) + \dots + b_m(x) = 1 \quad (\text{lineary dependent})$$

The fact that they are lineary dependent is a problem, we could prevent this by throwing out one of the binary factors, but if the test set gains a new category, it might be encoded by a zero binary factor, and with the deletion of which it would loose meaning.

Suppose we are trying to make a model for estimating the price of apartments using a linear model, where the categorical factor is the location of the apartment building. After performing one-hot encoding we would obtain the following result:

$$a(x) = w_1[f(x) = c_1] + \dots + w_m[f(x) = c_m] + \{\text{relation with other factors}\}$$

Which makes more sense than a standard linear model, as the location would define an “initial estimate” for the value of the apartment building, and only after all of the remaining factors would further adjust it.

Working with texts (bag of words).

Let’s borrow the same example from before, and suppose we are trying try predict its price via its textual description.

Find all of the words in our set: $\{c_1, \dots, c_m\}$

Format the text using m factors: $b_1(x), \dots, b_m(x)$, s.t. $b_{j(x)} := \text{num of words } c_j \text{ in the text.}$

The linear model would look like: $\alpha(x) = w_1 b_1(x) + \dots + w_m b_m(x) + \dots$

Essentially, each inclusion of the word c_j would change the predicted price by w_j .

Binarization of numerical factors.

Suppose that for the previous example we are given the distance to the metro (x_j). It seems clear that it there musn’t necesarrily be a linear dependency between the factor and the target variable.

To solve this problem we can binarize the factors.

For this we take a $\{t_1, \dots, t_m\}$ net, and add the points $t_0 = -\infty, t_{m+1} = +\infty$.

Set up new factors as: $b_{i(x)} = [t_{i-1} < x_j \leq t_i]$, $i = 1, \dots, m + 1$

The linear model would look like: $\alpha(x) = w_1 [t_0 < x_j \leq t_1] + \dots + w_{m+1} [t_m < x_j \leq t_{m+1}]$

Thus we have considred the non linear dependency between the factor and the target variable when predicting the price of the apartment for each interval of distance from the nearest metro.

2.3. Estimating errors in regression models

Previously we’ve spoken that there is a need to properly evaluate the quality of the predictions given by a model. Suppose that a is the model prediction, and y is our target variable. Here we will look through several methods of evaluating the deviation of the prediction from the original answer, that we will call $L(y, a)$.

MSE and R^2 .

The main way to evaluate the deviation of the prediction (answer) of the model a , and the target variable y .

$$L(y, a) = (a - y)^2$$

Mean Squared Error (MSE)

The main error function based on the squared difference.

$$\text{MSE}(a, X) := \frac{1}{\ell} \sum_{i=1}^{\ell} (\alpha(x_i) - y_i)^2$$

As all things, this error function is not perfect, as it's results are not great for interpretation, as it does not keep the unit of measurement of the original target variable. To avoid this problem the root squared error is introduced (RMSE).

$$\text{RMSE}(a, X) := \sqrt{\frac{1}{\ell} \sum_{i=1}^{\ell} (\alpha(x_i) - y_i)^2}$$

RMSE and MSE are good for comparing two different models, for quality control when training models, but does not allow us to derive conclusions on how good our model solves a particular task.

MSE = 10, would be an awful result if the target variable y would take values (0,1). However it would be a great result if y would take values (10000, 100000).

Coefficient of determination R^2 .

Essentially, the coefficient of determination R^2 is a normalized MSE, the closer it is to 1 the better the model is at making predictions, and the closer it is to 0, the closer it is in quality to having a constant prediction.

$$R^2(a, X) = 1 - \frac{\sum_{i=1}^{\ell} (a(x_i) - y_i)^2}{\sum_{i=1}^{\ell} (y_i - \bar{y})^2}, \quad \bar{y} = \frac{1}{\ell} \sum_{i=1}^{\ell} y_i$$

Mean absolute error (MAE).

Suppose we've replaced the squared difference with a module.

$$L(y, a) = |a - y|$$

The error function resulting from this change is the mean absolute error (MAE)

$$\text{MAE}(a, X) := \frac{1}{\ell} \sum_{i=1}^{\ell} |a(x_i) - y_i|$$

The mean absolute error (MAE) is not differentiable but is less sensitive to outliers that might ruin a model in more outlier sensitive error functions such as MSE, RMSE, and coefficient of determination.

Let's show why the mean absolute error (MAE) is less sensitive to outliers.

Suppose all ℓ objects from $\mathbb{X}$ have the same factor descriptions, but different values of the target variable $y_1, \dots, y_\ell$. In this case the model has to give the same answer on all these $\mathbb{X}$. If we picked MSE as the error function, we get the following problem⁶:

$$\frac{1}{\ell} \sum_{i=1}^{\ell} (\alpha(x_i) - y_i)^2 \rightarrow \min_a$$

The minimum would be obtained on the average of all the answers:

$$a_{\text{MSE}}^* = \frac{1}{\ell} \sum_{i=1}^{\ell} y_i$$

3. Linear Regression

4. Linear Classification

5. Linear Classification 2: Electric Boogaloo

5.0.1. Этапы решения задачи в ML.

1. Постановка задачи.

2.1. Сбор данных

2.2. Разработка признаков.

2.3 Подготовка данных.

3. Выбор функции потерь.

3

4

5

5.0.2. Линейная Классификация

$\mathbb{Y} = \{1, \dots, k\}$ = многоклассовая классификация

$$\mathbb{Y} = \left\{ \underbrace{-1}_{\text{отриц классиф.}}, \underbrace{+1}_{\text{отриц класс}} \right\} - \text{бинарная классификация}$$

$$\alpha(x) := \text{sign} \langle w, x \rangle$$

Что делать если $\langle w, x \rangle = 0$?

1. фиксированный класс выдавать
2. случайный класс
3. сказать что такого не бывает
4. отказ от классификации

$$\alpha(x) := \text{sign} (\langle w, x \rangle - t), \quad t - \text{гиперparameter}$$

Знаем:

$\langle w, x \rangle$ — условие ортогональности, и является уравнением гиперплоскости, w — вектор нормали

there should be a reminder from linear algebra here, a graph and stuff (add later)

5.0.3. Обучение

Запишем функционал:

$$Q(\alpha, X) = \frac{1}{\ell} \sum_{i=1}^{\ell} [\alpha(x_i) \neq y_i] - \text{доля ошибок (error rate)}$$

Проблема тут что производная либо 0 либо отсутствует.

$$\frac{1}{\ell} \sum_{i=1}^{\ell} [\text{sign} \langle w, x_i \rangle \neq y_i] = \frac{1}{\ell} \sum_{i=1}^{\ell} \left[\underbrace{y_i \langle w, x_i \rangle}_{M_i} < 0 \right] \rightarrow \min_w$$

Margin.

$$M_i = y_i \langle w, x_i \rangle - \text{отступ (margin)}$$

Где $\text{sign } M$ говорит о корректности классификации, и $|M|$ говорит об уверенности модели.

Пока что наша функция потерь:

$L(M) = [M < 0]$ — пороговая ф.п.

make a graph a bit later here

Идея:

$$[\mu < 0] \leq L^\sim(\mu) \text{ — дифф верная оценка}$$

$$0 \leq \frac{1}{\ell} \sum_{i=1}^{\ell} [y_i \langle w, x_i \rangle < 0] \leq \frac{1}{\ell} \sum_{i=1}^{\ell} L^\sim(y_i \langle, x_i \rangle) \rightarrow \min_w$$

Можно попробовать по подбирать разные:

1. $L_1^\sim(\mu) = \log(1 + \exp(-\mu))$ log ф.п
2. $L_2^\sim(\mu) = \max(0, 1 - \mu)$ hinge loss
3. $L_3^\sim(\mu) = \frac{\arctan(-\mu)}{\pi} + 1$

Далее будем рассматривать как выбирать между верхней оценки, но пока просто представим что мы некую выбрали.

$$\frac{1}{\ell} \sum_{i=1}^{\ell} L^\sim(y_i \langle w, x_i \rangle) + \alpha R(w) \rightarrow \min_w$$

6. Метрики качества классификации

6.0.1. Доля верных ответов. (Ассигасу)

$$\frac{1}{\ell} \sum_{i=1}^{\ell} [\alpha(x_i) = y_i]$$

Какая тут есть проблема?

Дизбаланс классов.

class sign	num	res
+1 :	50	$\alpha(x) = -1$
-1 :	950	ассигасу = 0.95

Мораль: Когда работаете с задачей классификации всегда необходимо смотреть на баланс классов.

Далее, когда мы улучшаем модель, есть два способа измерения прироста:

1. $|r_1 - r_2|$
2. $|r_1 - r_2| r_1$

Рассмотрим пример:

r_1, r_2	abs	rel
20%, 10%	10%	50%
50%, 25%	25%	50%
0.1%, 0.01%	0.09%	90%

7. Многоклассовая классификация и категориальные признаки.

Матрица ошибок.

$$\alpha(x) = +1, \quad \alpha(x) = -1$$

		$y = +1$	$y = -1$
Worked	$\alpha(x) = +1$	"True positive (TP)"	"False positive (FP)"
Did not work	$\alpha(x) = -1$	"False negative (FN)"	"True negative (TN)"

Accuracy

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Хорошо будет работать для балансированных классов (50-50). В противном случае делать нечего.

Precision. (точность)

Насколько мы доверяем модели.

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Замечание: Будем говорить что $\frac{0}{0} = 1$

Можно работать с несбалансированными классами.

Recall. (полнота).

Насколько модель покрывает положительный класс.

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Допустим:

$$a(x) = +1 > y = +1 \Rightarrow \text{precision} \uparrow, \text{recall} \downarrow$$

$$a(x) = +1 < y = +1 \Rightarrow \text{precision} \downarrow, \text{recall} \uparrow$$

Как можно балансировать между precision & recall?

Вспомним что модель классификации выглядит как:

$$\alpha(x) = \text{sign} \left(\underbrace{b(x)}_{\text{уверенность}} - \underbrace{t}_{\text{порог}} \right)$$

Шаг 1.

Отсортируем X по $b(x)$.

$$b(x) = 105, 99, \overset{t}{|}, \dots, 81, 73, \dots, -50, -70 \Rightarrow \text{precision} \uparrow, \text{recall} \downarrow$$

$$b(x) = 105, 99, \dots, 81, 73, \dots, \overset{t}{|}, -50, -70 \Rightarrow \text{precision} \downarrow, \text{recall} \uparrow$$

Обычно precision & recall балансируются выбором порога.

$$\begin{cases} \text{precision} \rightarrow \max \\ \text{recall} \geq 0.9 \end{cases}, \quad \begin{cases} \text{recall} \rightarrow \max \\ \text{precision} \geq 0.95 \end{cases}$$

Как объединить точность (precision) и полноту (accuracy)?

1. Среднее арифметическое.

Почему это нам не подойдет?

$$\text{pr} = 0.05, \text{rec} = 0.9 \Rightarrow A = 0.475, \text{ but pr}$$

2. Минимум.

$$M = \min(\text{pr}, \text{rec})$$

$\text{pr} = 0.5, \text{rec} = 0.5$ and $\text{pr} = 1, \text{rec} = 0.5$ but this model would give them same eff

3. Среднее гармоническое. (F-мера)

$$F := \frac{2\text{pr} \cdot \text{rec}}{\text{pr} + \text{rec}}$$

3.2 (F_α -мера)

$$F_\alpha = \frac{(1 + \alpha^2) \cdot \text{pr} \cdot \text{rec}}{\alpha^2 \cdot \text{pr} + \text{rec}} \quad \begin{cases} \xrightarrow{\alpha \rightarrow 0} \text{pr} \\ \xrightarrow{\alpha \rightarrow +\infty} \text{rec} \end{cases}$$

4. Среднее геометрическое.

$$G = \sqrt{\text{pr} \cdot \text{rec}} - \text{ближе к ариф.}$$

$$\text{pr} = 0.9, \text{rec} = 0.1 \Rightarrow$$

$$F < G$$

Отступление.

$$\text{lift} = \frac{\text{pr}}{\frac{\text{TP} + \text{FN}}{\ell}} = \frac{\overline{\text{pr}}}{\frac{\ell_+}{\ell}} \quad (1)$$

(1) - доля полезных звонков если звоноили случайно (2)

В итоге,

$$\text{lift} = 1 \Rightarrow \text{модель рандомна} \quad \text{lift} < 1 \Rightarrow \text{worse than random}$$

5. Площади под кривыми (Метрики качества ранжирования).

b(x)	y	y
100	+1	+1
93	+1	-1
91	+1	+1
...	...	...
-1	-1	+1
-2	-1	-1
-10	-1	+1
	good	bad

5.1 ROC (Reversed operated characteristic) - кривая.

		y = +1	y = -1
Worked	$\alpha(x) = +1$	"True positive (TP)"	"False positive (FP)"
Did not work	$\alpha(x) = -1$	"False negative (FN)"	"True negative (TN)"

False positive rate.

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} = \frac{\text{FP}}{\ell_-}$$

True positive rate.

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{TP}}{\ell_+}$$

Далее перебираем все t , рисуем что-то что-то что-то.

Пример 1: Модель которая сначала ставит все положительные а потом все отрицательные.

b(x)	y
	+1
	+1
	-1
	-1

$$\text{AUC} - \text{ROC} = 1$$

Пример 2:

Рассмотрим модель которая чередует объекты.

b(x)	y
	+1
	-1
	+1
	-1

$$\text{AUC} - \text{ROC} = \frac{1}{2}$$

Пример 3: Модель которая сначала ставит все отрицательные а потом все положительный.

b(x)	y
	-1
	-1
	+1

$$\text{AUC} - \text{ROC} = 0$$

Показывает насколько полезная модель оказывается.

Рассмотрим случай:

b(x)	y
0.7	+1
0.7	-1
0.7	-1

5.2 Precision-Recall (PR) кривая.

AUC - ROC, чем больше тем лучше.

Замечание 1. Индекс Джини.

$$\text{Gini} = 2 \cdot \text{AUC} \cdot \text{ROC} - 1 = 2 \left(\text{AUC} \cdot \text{ROC} - \frac{1}{2} \right)$$

Замечание 2.

AUC – ROC может вводить в заблуждение если в выборке много легких (объекты с $\mu > 0$ примеров).

b(x)	
-1	EDA
-1	EDA
+1	AMI
+1	AMI
+1	AMI
-1	Other
-1	Other

AUC-ROC не очень чувствительный к небольшому количеству серьезных FP ошибок.

Как это решить?

1. AUC-ROC @ k
2. AUC - PR

8. Функция потерь в классификации.

$$\frac{1}{\ell} \sum_{i=1}^{\ell} (\text{sign } \langle w, x_i \rangle \neq y_i) \leq \frac{1}{\ell} \sum_{i=1}^{\ell} L()$$